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Do your level best

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Editor's Note: There is a figure that accompanies this story that can be downloaded in PDF format via the "Download Now" button at the bottom of the page.

Plant engineers who complain about the difficulty of getting good level measurements have plenty of company. A recent poll of readers of *Control* magazine showed that level measurement is considered one of the two most vexing problems by process automation professionals. Figure 1 summarizes the results of another *Control* poll about the reasons for the problems. Two factors clearly garnered the most responses: The application was poorly designed, and the level of the fluid or solid is very hard to measure.

Many options are available for measuring a level, including simple bubbler and mechanical sensors, differential-pressure devices and ultrasonic- and radar-based units. As the figure indicates, however, none of these escaped brickbats from *Control* readers when asked which technology gave them the most problems.

Part of the problem is that many process engineers do not think of level as a control function. They treat temperature and pressure as critical for most unit operations and only consider level as a side issue. Many process simulator programs also pay scant attention to level measurement, and conventional tuning software isn't always optimized for level loops.

In addition, level measurement often suffers neglect from plant staff because it can be difficult. For instance, a stirred or sparged reactor might contain a vortex or froth, both of which can make determining the equivalent static level a tricky inferential process. Plus, in scale-up, controls are sometimes an afterthought and are not well integrated into the plant design.

Vessels are generally designed without ease of level measurement in mind. You find vessels built through floors of plants, those with only inlets and outlets and no measurement ports, and still others with so many internals that there are few ways to measure level. If the engineer who specs the vessel does not consider its place in the process control loop, the vessel likely will be difficult to instrument for level measurement.

A level-headed approach

Before specifying the vessel, the engineer should know the purpose of the vessel as well as the characteristics of the material it will contain and the measurement technology being proposed. But, in the end, it is the performance of the vessel and sensor within the entire process loop that matters. As John Gerry of software vendor ExperTune, Hubertus, Wis., says, "The performance goal of one level loop may be the opposite of another."

Some vendors, such as Endress+Hauser, Greenwood, Ind., offer online selection wizards for level-measurement technology. These wizards, which are application specific and not product-centric, are especially helpful for the engineer who only sporadically chooses such sensors. In addition, the level measurement section of the latest (fourth) edition of the "Instrument Engineer's Handbook" should be required reading for process and control system engineers; in it, Béla Lipták devotes more than 160 pages to the topic.

Knowing the physical characteristics of the material to be measured is crucial. One vendor, Magnetrol International, Downers Grove, Ill., provides a noncommercial "Technical Handbook" with selected physical properties, including dielectric constants, densities, viscosities, specific gravities and much more (Figure 2). This book, or one like it, is invaluable.

Armed with such information, you should be in a position to start sorting through level-

measurement options. In the case of clean fluids, you probably can use anything you want, from a float-type gauge to a person with a stick. Even simple devices like float gauges can talk to distributed control systems. For instance, K-Tek, Prairieville, La., offers the AT200 transmitter with HART interface. If you need a noncontacting device, the choice can get more complicated. An ultrasonic gauge might do the trick. However, if you are using a pressure vessel or the vessel contains foam or large amounts of vapor or dust, you should go with noncontact radar, presuming the sensor can be installed in an appropriate location. If not, then you likely are limited to a nuclear gauge or, perhaps, a load cell.

If you need high accuracy, there is a sensor to fit nearly any vessel and all media — from differential pressure (Emerson Rosemount, Foxboro, Honeywell, etc.) and tank-farm radar (Emerson Saab-Rosemount, Endress+Hauser), to hydrostatic tank gauging systems (King Gauge, Varec) and load cells (Hardy, Mettler Toledo, etc.).

Paying adequate attention to level will help keep your process on an even kilter.

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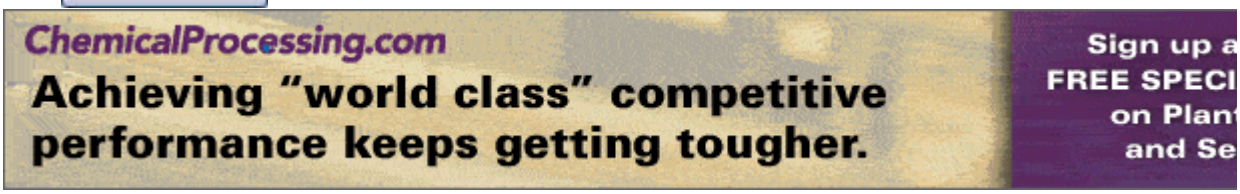

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